

Module / port diagram

Every RTL block starts with a boundary you can draw on paper

Modules, ports, and instances

- A module is a named design unit
- Ports are the named wires on its edge, input, output, or inout
- Inside the module you describe behavior with assigns or always blocks
- Named connects like dot-a open-paren a close-paren tie a formal port to an actual wire
- Hierarchy is simply the tree of parents and instances you get when one module contains

Browser lab

igital Design and Verification Platform

Home

Tools

Module diagram

Challenges 0/22

PICK ONE

Quiz: module

A Verilog module...

☐ defines a named design unit with ports (and optional internals)

☐ is only a \$finish call

☐ must be analog

☐ cannot be instantiated

Start

Show hint

Check

Next

Stop

Idle

Source

and2 (starter)

One module, two inputs, one output — classic port picture.

Counter ports

clk / rst_n / enable / q[3:0] — mixed directions & a bus.

Two instances

top wires two and2 gates — hierarchy list grows.

top = and2 instance

Parent instantiates child with named port connects.

Parameterized slice

R[WIDTH-1:0] on an instance — parameters = ports.

```
module and2(  
  input a,  
  input b,  
  output y  
);  
  assign y = a & b;  
endmodule
```

Parse / redraw

Reset to preset text

Parsed 1 module(s).

Uses the HDL engine parser API — teaching subset, not a full LRM front-end.

Hierarchy

VIEW MODULE

and2

and2 (3 ports, 0 inst)

Port diagram

and2

INPUTS

a

INPUT

b

INPUT

OUTPUTS / OTHER

y

OUTPUT

MODULE

Port detail

Click a port pin to see its direction and role.

Ready — click ports or load a hierarchy preset.

Module diagram lab starter

Real Verilog practice

```
wsl - module01-module-diagram/examples/navigation

# Track A - real Unix
# real Verilog session (Track A)

$ cat examples/and2/and2.v
module and2(
    input  a,
    input  b,
    output y
);
    assign y = a & b;
endmodule

$ iverilog -t null examples/and2/and2.v
(syntax check passed)
```

Pitfalls to watch

- Do not swap input and output directions, a port driven inside must be an output
- Do not assume positional port order when named connects are clearer
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog, write or edit a minimal module shell for this idea
- When you are ready